EXP - 02.

DATE: 10/07/26

DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add constraints like **PRIMARY KEY, FOREIGN KEY, UNIQUE KEY, and CHECK** using DDL commands in the **Saveetha Hospital** database.

Description:

- **PRIMARY KEY**: Ensures each record in a table is uniquely identified.
- **FOREIGN KEY**: Links two tables together, referencing the primary key of another table.
- **UNIQUE**: Ensures all values in a column are different.
- **CHECK**: Restricts values in a column to a defined condition.

Questions (Saveetha Hospital Context)

1. **Alter the table PATIENT with following structure**

○ Column: PatientID → **PRIMARY KEY**
○
sql

ALTER TABLE Patient ADD PRIMARY KEY(PatientID);

2. **Alter the table FACULTY with following structure**

○ Column: FacNo → **PRIMARY KEY**
○ Column: Gender → **CHECK (‘M’ or ‘F’)**
sql

ALTER TABLE Faculty ADD PRIMARY KEY(FacNo);

ALTER TABLE Faculty ADD CONSTRAINT chk_gender CHECK (Gender IN ('M','F'));

3. **Alter the table DEPARTMENT with following structure**

○ Column: DeptNo → **PRIMARY KEY**

sql

ALTER TABLE Department ADD PRIMARY KEY(DeptNo);

4. **Alter the table COURSE with following structure**

○ Column: CourseNo → **PRIMARY KEY**
○ Column: SemNo → **CHECK (SemNo BETWEEN 1 AND 6)**

sql

ALTER TABLE Course ADD PRIMARY KEY(CourseNo);

ALTER TABLE Course ADD CONSTRAINT chk_sem CHECK (SemNo BETWEEN 1 AND 6);

5. **Alter the table PATIENT with following structure**

○ Column: PatientName → **UNIQUE KEY**

sql

ALTER TABLE Patient ADD CONSTRAINT unique_patient UNIQUE(PatientName);

6.) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

OUTPUTS:

1.

```
mysql> ALTER TABLE student3
    -> ADD CONSTRAINT fk_regno FOREIGN KEY(RegNo)
    -> REFERENCES student2(RegNo)
    -> ON DELETE CASCADE
    -> ON UPDATE CASCADE;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC student3;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | NO   | MUL | NULL    |       |
| SName    | varchar(15) | YES  |     | NULL    |       |
| Age      | char(2)     | YES  |     | NULL    |       |
| MobileNo | int         | YES  |     | NULL    |       |
| Address  | varchar(15) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

2.

```
mysql> SHOW CREATE TABLE student3;
+----------+------------------------------------------------------------+
| Table    | Create Table                                               |
+----------+------------------------------------------------------------+
| student3 | CREATE TABLE `student3` (                                  |
|          |   `RegNo` int NOT NULL,                                    |
|          |   `SName` varchar(15) DEFAULT NULL,                        |
|          |   `Age` char(2) DEFAULT NULL,                              |
|          |   `MobileNo` int DEFAULT NULL,                             |
|          |   `Address` varchar(15) DEFAULT NULL,                      |
|          |   CONSTRAINT `fk_regno` FOREIGN KEY (`RegNo`)              |
|          |   REFERENCES `student2` (`RegNo`) ON DELETE CASCADE        |
|          |   ON UPDATE CASCADE                                        |
|          | ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4                    |
+----------+------------------------------------------------------------+
1 row in set (0.00 sec)
```

3.

```
mysql> ALTER TABLE department ADD PRIMARY KEY(deptno);
Query OK, 0 rows affected (0.11 sec)

mysql> DESC department;
+----------+----------+------+-----+---------+-------+
| Field    | Type     | Null | Key | Default | Extra |
+----------+----------+------+-----+---------+-------+
| deptno   | int(10)  | NO   | PRI | NULL    |       |
| deptname | char(10)| YES  |     | NULL    |       |
| location | char(10)| YES  |     | NULL    |       |
+----------+----------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

4.

```
mysql> ALTER TABLE course ADD PRIMARY KEY(courseno);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> ALTER TABLE course ADD CHECK(semno>=1 && semno<=6);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC course;
+-------------+---------+------+-----+---------+-------+
| Field       | Type    | Null | Key | Default | Extra |
+-------------+---------+------+-----+---------+-------+
| courseno    | int(10) | NO   | PRI | NULL    |       |
| coursedesc  | char(15)| YES  |     | NULL    |       |
| coursetype  | char(15)| YES  |     | NULL    |       |
| semno       | int(10) | YES  |     | NULL    |       |
| hallno      | int(10) | YES  |     | NULL    |       |
| facno       | int(10) | YES  |     | NULL    |       |
+-------------+---------+------+-----+---------+-------+
6 rows in set (0.00 sec)
```

5.

```
mysql> ALTER TABLE patient ADD UNIQUE KEY(PName);
Query OK, 0 rows affected (0.07 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> DESC patient;
+---------+-------------+------+-----+---------+-------+
| Field   | Type        | Null | Key | Default | Extra |
+---------+-------------+------+-----+---------+-------+
| RegNo   | int         | NO   | PRI | NULL    |       |
| PName   | char(15)    | YES  | UNI | NULL    |       |
| Age     | char(2)     | YES  |     | NULL    |       |
| Address | varchar(15) | YES  |     | NULL    |       |
+---------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

6.

```
mysql> DESC hospital;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CourseNo   | varchar(3)  | NO   | PRI | NULL    |       |
| CourseDesc| varchar(14) | YES  |     | NULL    |       |
| CourseType| char(1)     | YES  |     | NULL    |       |
| SemNo      | char(1)     | YES  |     | NULL    |       |
| HallNo     | varchar(4)  | YES  |     | NULL    |       |
| FacNo      | varchar(4)  | YES  |     | NULL    |       |
| Dept       | varchar(15) | NO   |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

RESULT:

This is the concise confirmation line MySQL shows after successfully executing the ALTER TABLE command.